

Tao Lin

✉ lintao@cuhk.edu.cn
🌐 <https://tao-l.github.io/>

Education

- **Harvard University**, Ph.D. in Computer Science 2020/09 – 2025/05
Advisor: *Yiling Chen*
Dissertation: Incentive Design in the Machine Learning Age
Areas of Research: Economics and Computation, Mechanism Design, Information Design, Game Theory, Machine Learning, Theoretical Computer Science
- **Peking University**, B.Sc. in Computer Science and Technology (*summa cum laude*) 2016/09 – 2020/05
Advisor: *Xiaotie Deng*
Thesis: Private Information Protection Game in Auctions

Experience

- **The Chinese University of Hong Kong, Shenzhen**, School of Data Science 2026/07 – Now
Assistant Professor (in Computer Science)
- **Microsoft**, Postdoctoral Researcher 2025/07 – 2026/06
Host: *Aleksandrs Slivkins*
- **Google**, Student Researcher 2024, 06 – 09
Mentor: *Christopher Liaw*
- **ByteDance**, Research Intern 2023, 05 – 09
Mentor: *Yang Liu*
- **University of British Columbia**, Short-term Research Visit 2019, 07 – 09
Mentor: *Hu Fu*

Research Interests

My research focuses on **Learning-Based Incentive Design**, an interdisciplinary topic in economics, machine learning, and theoretical computer science. A central question I study is:

*How to design the incentive structure of a multi-agent system
so that self-interested agents autonomously reach desired outcomes?*

Specific directions include *multi-agent learning theory*, *mechanism design*, *information design*, *learning in games*, *incentives in machine learning systems*, and *algorithmic game theory*.

My research is often motivated by the interplay between economic incentives and machine learning algorithms in real-world AI systems, such as *advertising auctions*, *recommender systems*, and *large language models (LLM)*.

Conference Publications

- [Explainable Information Design](#) [EC 2026]
(alphabetical) Yiling Chen, *Tao Lin*, Wei Tang, Jamie Tucker-Foltz
- [Information Design With Large Language Models](#) [EC 2026]
(alphabetical) Paul Duetting, Safwan Hossain, *Tao Lin*, Renato Paes Leme, Sai Srivatsa Ravindranath, Haifeng Xu, Song Zuo

- [Gradient Dynamics in First-Price Auctions: Iterative Strategy Elimination via Cubic Potentials](#) [EC 2026]
Mete Şeref Ahunbay, Weiqiang Zheng, *Tao Lin*
- [On the Coordination of Value-Maximizing Bidders](#) [ICML 2026]
Yanru Guan, Jiahao Zhang, Zhe Feng, *Tao Lin*
- [Reward Shaping for Inference-Time Alignment: A Stackelberg Game Perspective](#) [ICML 2026]
Haichuan Wang, *Tao Lin*, Lingkai Kong, Ce Li, Hezi Jiang, Milind Tambe
- [Learning a Game by Paying the Agents](#) [ICLR 2026]
Brian Hu Zhang*, *Tao Lin**, Yiling Chen, Tuomas Sandholm
- [Learning to Play Multi-Follower Bayesian Stackelberg Games](#) [ICLR 2026]
Gerson Personnat, *Tao Lin*, Safwan Hossain, David C. Parkes
- [A Unified Approach to Submodular Maximization Under Noise](#) [NeurIPS 2025]
(alphabetical) Kshipra Bhawalkar, Yang Cai, Zhe Feng, Christopher Liaw, *Tao Lin*
- [Generalized Principal-Agent Problem with a Learning Agent](#) [ICLR 2025]
Tao Lin, Yiling Chen (spotlight)
- [Information Design with Unknown Prior](#) [ITCS 2025]
Tao Lin, Ce Li
- [User-Creator Feature Polarization in Recommender Systems with Dual Influence](#) [NeurIPS 2024]
Tao Lin, Kun Jin, Andrew Estornell, Xiaoying Zhang, Yiling Chen, Yang Liu
- [Bias Detection via Signaling](#) [NeurIPS 2024]
(alphabetical) Yiling Chen, *Tao Lin*, Ariel D. Procaccia, Aaditya Ramdas, Itai Shapira
- [Multi-Sender Persuasion: A Computational Perspective](#) [ICML 2024]
Safwan Hossain*, Tonghan Wang*, *Tao Lin**, Yiling Chen, David C. Parkes, Haifeng Xu
- [Learning Thresholds with Latent Values and Censored Feedback](#) [ICLR 2024]
Jiahao Zhang, *Tao Lin*, Weiqiang Zheng, Zhe Feng, Yifeng Teng, Xiaotie Deng
- [Sample Complexity of Forecast Aggregation](#) [NeurIPS 2023]
Tao Lin, Yiling Chen (spotlight)
- [From Monopoly to Competition: Optimal Contests Prevail](#) [AAAI 2023]
(alphabetical) Xiaotie Deng, Yotam Gafni, Ron Lavi, *Tao Lin*, Hongyi Ling
- [Nash Convergence of Mean-Based Learning Algorithms in First Price Auctions](#) [WWW 2022]
(alphabetical) Xiaotie Deng, Xinyan Hu, *Tao Lin*, Weiqiang Zheng
- [How Many Representatives Do We Need? The Optimal Size of a Congress Voting on Binary Issues](#) [AAAI 2022]
Manon Revel, *Tao Lin*, Daniel Halpern
- [Learning Utilities and Equilibria in Non-Truthful Auctions](#) [NeurIPS 2020]
(alphabetical) Hu Fu, *Tao Lin*
- [A Game-Theoretic Analysis of the Empirical Revenue Maximization Algorithm with Endogenous Sampling](#) [NeurIPS 2020]
(alphabetical) Xiaotie Deng, Ron Lavi, *Tao Lin*, Qi Qi, Wenwei Wang, Xiang Yan
- [Private Data Manipulation in Optimal Sponsored Search Auction](#) [WWW 2020]
(alphabetical) Xiaotie Deng, *Tao Lin*, Tao Xiao

Journal Publications

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- **Generalized Principal-Agent Problem with a Learning Agent** [Quantitative Economics
Tao Lin, Yiling Chen (forthcoming)]

- [Information Design with Large Language Models: An Annotated Reading List](#) [ACM SIGecom Exchanges, 2026]
Tao Lin, Safwan Hossain, Yiling Chen
- [From Monopoly to Competition: When do Optimal Contests Prevail?](#) [Games and Economic Behavior, 2025]
(alphabetical) Xiaotie Deng, Yotam Gafni, Ron Lavi, *Tao Lin*, Hongyi Ling
- [Private Data Manipulation in Sponsored Search Auctions](#) [CAAI Artificial Intelligence Research, 2023]
(alphabetical) Xiaotie Deng, *Tao Lin*, Tao Xiao

Notes Not Planned to Publish

- [How Does Independence Help Generalization? Sample Complexity of ERM on Product Distributions](#) [2022]
- [On Clearing Prices in Matching Markets: A Simple Characterization without Duality](#) [2019]

Teaching Experiences

- *Instructor* for **CSC6011 Theory of Computation** (CUHKSZ) Fall 2026
- *Teaching assistant* for **Convex Optimization and Its Applications** (Harvard University) Spring 2022
- *Teaching assistant* for **Algorithmic Game Theory** (Peking University) Fall 2019

Academic Services

- Organizer of EC'25 Workshop on "Information Economics X LLMs" 2025
- Organizer of Harvard EconCS seminar 2023 – 2024
- Conference Review: NeurIPS, ICML, ICLR, AAAI, AISTATS, ACML, PPAI, STOC, FOCS, SODA, ITCS, COLT, IJTCS, EC, WINE, SAGT, CIST
- Journal Review: SIAM Journal on Computing, Theoretical Computer Science, Journal of Artificial Intelligence Research, Transactions of Machine Learning Research, ACM Transactions on Economics and Computation

Selected Talks (Excluding Conference Paper Presentations)

- [Information Design with Large Language Models](#)
(2026/09) CCF 计算经济学年会
(2026/06) Carnegie Mellon University
(2026/04) Boston Economics and Computing Hub (BEACH) Workshop
(2026/01) POMS-HK International Conference
- Explainable Information Design
(2026/06) Talk at Northwestern University
- Reward Shaping for (Inference-Time) Alignment
(2026/06) Talk at University of Chicago
- Non-Bayesian Information Design: Learning and LLM-Based Approaches
(2026/01) Talks at SHUFE, SJTU, and Huawei

- Learning to Coordinate Bidders in Non-Truthful Auctions
(2025/10) INFORMS Annual Meeting
- How to Avoid Polarization in Recommender Systems with Dual Influence?
(2024/11) Chinese University of Hong Kong, Computer Science and Engineering Seminar
- Bayesian Persuasion with a Learning Agent
(2024/10) INFORMS Annual Meeting
- Generalized Principal-Agent Problem with a Learning Agent
(2024/08) ESIF Economics and AI+ML Meeting
- Private Data Manipulation in Sponsored Search Auctions
(2023/08) Invited talk at CCF 计算经济学年会
- [Sample Complexity of Forecast Aggregation](#)
(2023/06) Peking University Turing Class “CS peer talk”
- Persuading a Behavioral Agent: Approximately Best Responding and Learning
(2023/03) Harvard EconCS seminar
- [Nash Convergence of Mean-Based Learning Algorithms in First Price Auctions](#)
(2022/05) Invited talk at AAMAS Workshop on Learning with Strategic Agents
- Robustness of Empirical Revenue Maximization in Auction Learning
(2020/12) Institute for Theoretical Computer Science (ITCS), SHUFE

Awards

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| - Siebel Scholarship
(Annually awarded for academic excellence and demonstrated leadership to 80 top students from the world’s leading graduate schools.) | 2024 |
| - Peking University Turing Class Scholarship | 2019 |
| - Peking University “Fang Zheng” Scholarship | 2017 |
| - Chinese National Olympiad in Informatics, Silver Medal | 2015 |

References

Yiling Chen

Gordon McKay Professor of Computer Science
John A. Paulson School of Engineering and Applied Sciences
Harvard University
yiling@seas.harvard.edu

Ariel D. Procaccia

Gordon McKay Professor of Computer Science
John A. Paulson School of Engineering and Applied Sciences
Harvard University
arielpro@seas.harvard.edu

Ron Lavi

Reader (Associate Professor), Department of Economics
University of Bath
arl65@bath.ac.uk

Haifeng Xu

Associate Professor, Department of Computer Science and Data Science Institute
University of Chicago

haifengxu@uchicago.edu